20260914 Quiz

for (MATH210) Applied Complex Variables

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Exercise 3.1. Reduce each of these quantities to a real number:

(a) $\dfrac{1+2i}{3-4i}+\dfrac{2-i}{5i}$.

(b) $\dfrac{5i}{(1-i)(2-i)(3-i)}$.

(c) $(1-i)^4$.

Solution.

(a) Multiplying the first denominator by its conjugate and simplifying the second term,

$$\frac{1+2i}{3-4i}=\frac{(1+2i)(3+4i)}{25}=-\frac{1}{5}+\frac{2}{5}i,$$
$$\frac{2-i}{5i}=\frac{(2-i)(-i)}{5}=-\frac{1}{5}-\frac{2}{5}i.$$

Hence

$$\boxed{\frac{1+2i}{3-4i}+\frac{2-i}{5i}=-\frac{2}{5}}.$$

(b) Since

$$(1-i)(2-i)=1-3i$$

and

$$(1-3i)(3-i)=-10i,$$

we obtain

$$\boxed{\frac{5i}{(1-i)(2-i)(3-i)}=-\frac{1}{2}}.$$

(c) We have

$$(1-i)^2=-2i,$$

so

$$\boxed{(1-i)^4=(-2i)^2=-4}.$$

□

Exercise 5.4. Verify that

$$\sqrt{2}\,|z| \geq |\operatorname{Re} z| + |\operatorname{Im} z|.$$

Suggestion. Reduce this inequality to $(|x| - |y|)^2 \geq 0$.

Solution. Write $z = x + iy$. Both sides are nonnegative, so it is enough to compare their squares. The desired inequality is equivalent to

$$2(x^2 + y^2) \geq (|x| + |y|)^2,$$

which reduces to

$$x^2 + y^2 - 2|xy| = (|x| - |y|)^2 \geq 0.$$

Therefore

$$\boxed{\sqrt{2}\,|z| \geq |\operatorname{Re} z| + |\operatorname{Im} z|}.$$

□

Exercise 5.7. Show that for R sufficiently large, the polynomial $P(z)$ in Example 3, Sec. 5, satisfies the inequality

$$|P(z)| < 2|a_n||z|^n \qquad \text{whenever } |z| > R.$$

Suggestion. Observe that there is a positive number R such that the modulus of each quotient in inequality (9), Sec. 5, is less than $|a_n|/n$ when $|z| > R$.

Solution. As in Example 3, Sec. 5, write

$$P(z) = (a_n + w)z^n,$$

where

$$w = \frac{a_0}{z^n} + \frac{a_1}{z^{n-1}} + \cdots + \frac{a_{n-1}}{z}.$$

For sufficiently large R, each of the n terms on the right has modulus less than $|a_n|/n$ whenever $|z| > R$. Hence

$$|w| < |a_n|.$$

Using the triangle inequality,

$$|P(z)| = |a_n + w||z|^n \leq (|a_n| + |w|)|z|^n < 2|a_n||z|^n.$$

Thus the stated inequality holds for all $|z| > R$.

□

Exercise 6.7. Show that

$$\left|\operatorname{Re}\left(2 + \overline{z} + z^3\right)\right| \leq 4 \qquad \text{when } |z| \leq 1.$$

Solution. For every complex number w, $|\operatorname{Re} w| \leq |w|$. Hence, using the triangle inequality and $|\overline{z}| = |z|$,

$$\begin{aligned}\left|\operatorname{Re}\left(2 + \overline{z} + z^3\right)\right| &\leq \left|2 + \overline{z} + z^3\right| \\ &\leq 2 + |\overline{z}| + |z|^3 \\ &= 2 + |z| + |z|^3 \\ &\leq 4,\end{aligned}$$

since $|z| \leq 1$.

□

Exercise 6.13. Show that the equation $|z - z_0| = R$ of a circle, centered at z_0 with radius R, can be written

$$|z|^2 - 2\,\mathrm{Re}(z\overline{z_0}) + |z_0|^2 = R^2.$$

Solution. Squaring $|z - z_0| = R$ gives

$$\begin{aligned} R^2 &= |z - z_0|^2 \\ &= (z - z_0)(\overline{z} - \overline{z_0}) \\ &= |z|^2 - z\overline{z_0} - \overline{z}z_0 + |z_0|^2. \end{aligned}$$

The middle two terms are conjugates, so

$$z\overline{z_0} + \overline{z}z_0 = 2\,\mathrm{Re}(z\overline{z_0}).$$

Therefore

$$\boxed{|z|^2 - 2\,\mathrm{Re}(z\overline{z_0}) + |z_0|^2 = R^2}.$$

□

Exercise 9.9. Establish the identity

$$1 + z + z^2 + \cdots + z^n = \frac{1 - z^{n+1}}{1 - z} \qquad (z \neq 1)$$

and then use it to derive Lagrange's trigonometric identity:

$$1 + \cos\theta + \cos 2\theta + \cdots + \cos n\theta = \frac{1}{2} + \frac{\sin((2n+1)\theta/2)}{2\sin(\theta/2)} \qquad (0 < \theta < 2\pi).$$

Suggestion. As for the first identity, write $S = 1 + z + z^2 + \cdots + z^n$ and consider the difference $S - zS$. To derive the second identity, write $z = e^{i\theta}$ in the first one.

Solution. Let

$$S = 1 + z + z^2 + \cdots + z^n.$$

Then

$$S - zS = 1 - z^{n+1},$$

so, when $z \neq 1$,

$$\boxed{S = \frac{1 - z^{n+1}}{1 - z}}.$$

Now let $z = e^{i\theta}$, where $0 < \theta < 2\pi$. Then

$$\begin{aligned} 1 + e^{i\theta} + \cdots + e^{in\theta} &= \frac{1 - e^{i(n+1)\theta}}{1 - e^{i\theta}} \\ &= e^{in\theta/2}\frac{\sin((n+1)\theta/2)}{\sin(\theta/2)}. \end{aligned}$$

Taking real parts gives

$$1 + \cos\theta + \cdots + \cos n\theta = \frac{\sin((n+1)\theta/2)\cos(n\theta/2)}{\sin(\theta/2)}.$$

Using

$$2\sin\frac{(n+1)\theta}{2}\cos\frac{n\theta}{2} = \sin\frac{(2n+1)\theta}{2} + \sin\frac{\theta}{2},$$

we obtain

$$\boxed{1 + \cos\theta + \cdots + \cos n\theta = \frac{1}{2} + \frac{\sin((2n+1)\theta/2)}{2\sin(\theta/2)}}.$$

□

Exercise 11.6. Find the four zeros of the polynomial $z^4 + 4$, one of them being

$$z_0 = \sqrt{2}\, e^{i\pi/4} = 1 + i.$$

Then use those zeros to factor $z^2 + 4$ into quadratic factors with real coefficients.

Solution. The equation $z^4 + 4 = 0$ is

$$z^4 = 4e^{i(\pi+2k\pi)}.$$

Thus its four roots are

$$z = \sqrt{2}\, e^{i(\pi/4+k\pi/2)}, \qquad k = 0, 1, 2, 3,$$

namely

$$1 + i, \quad -1 + i, \quad -1 - i, \quad 1 - i.$$

Pairing conjugate roots gives real quadratic factors:

$$\begin{aligned} z^4 + 4 &= (z - (1 + i))(z - (1 - i))(z - (-1 + i))(z - (-1 - i)) \\ &= (z^2 - 2z + 2)(z^2 + 2z + 2). \end{aligned}$$

Hence

$$\boxed{z^4 + 4 = (z^2 + 2z + 2)(z^2 - 2z + 2)}.$$

□

Remark. The second polynomial in the printed statement is a typographical error. The four zeros requested are the zeros of $z^4 + 4$, and the printed answer factors $z^4 + 4$. Accordingly, the intended factorization is that of $z^4 + 4$.

Exercise 12.7. Determine the accumulation points of each of the following sets:

(a) $z_n = i^n$, $n = 1, 2, \ldots$.

(b) $z_n = i^n/n$, $n = 1, 2, \ldots$.

(c) $0 \le \arg z < \pi/2$, $z \neq 0$.

(d) $z_n = (-1)^n(1 + i)\dfrac{n - 1}{n}$, $n = 1, 2, \ldots$.

Solution.

(a) The set consists only of the four points $1, i, -1, -i$. A finite set has no accumulation points. Thus there are none.

(b) Since $|z_n| = 1/n \to 0$, every neighborhood of 0 contains points of the set other than 0. Conversely, fix $a \neq 0$ and choose N so large that $|z_n| < |a|/2$ for all $n > N$. A sufficiently small deleted neighborhood of a excludes these terms and also the finitely many distinct points among $z_1, \ldots, z_N$. Hence the only accumulation point is

$$\boxed{0}.$$

(c) The inequality in the statement specifies an argument of z in $[0, \pi/2)$; equivalently, $0 \le \operatorname{Arg} z < \pi/2$. Thus the set is the sector consisting of the nonzero points with angles from 0 up to, but not including, $\pi/2$. Every point of the closed first quadrant can be approached by points of this sector, and no point outside that closed quadrant can. Hence the set of accumulation points is

$$\boxed{\{x + iy \mid x \ge 0,\ y \ge 0\}}.$$

(d) For even n,

$$z_n = (1+i)\frac{n-1}{n} \longrightarrow 1+i,$$

while for odd n,

$$z_n = -(1+i)\frac{n-1}{n} \longrightarrow -(1+i).$$

These are the only two accumulation points, so

$$\boxed{\pm(1+i)}.$$

□

Exercise 14.2. In each case, write the function $f(z)$ in the form $f(z) = u(x,y) + iv(x,y)$:

(a) $f(z) = z^3 + z + 1$.

(b) $f(z) = \dfrac{\overline{z}^2}{z}$, $z \neq 0$.

Suggestion. In part (b), start by multiplying the numerator and denominator by $\overline{z}$.

Solution. Write $z = x + iy$.

(a) Expanding $(x+iy)^3$,

$$\begin{aligned} f(z) &= (x+iy)^3 + (x+iy) + 1 \\ &= (x^3 - 3xy^2 + x + 1) + i(3x^2y - y^3 + y). \end{aligned}$$

Therefore

$$\boxed{u = x^3 - 3xy^2 + x + 1, \qquad v = 3x^2y - y^3 + y}.$$

(b) Since $z \neq 0$,

$$\frac{\overline{z}^2}{z} = \frac{\overline{z}^3}{|z|^2}.$$

Now

$$\overline{z}^3 = (x-iy)^3 = (x^3 - 3xy^2) + i(y^3 - 3x^2y),$$

and hence

$$\boxed{f(z) = \frac{x^3 - 3xy^2}{x^2+y^2} + i\frac{y^3 - 3x^2y}{x^2+y^2}}.$$

□

Exercise 14.3. Suppose that

$$f(z) = x^2 - y^2 - 2y + i(2x - 2xy), \qquad z = x + iy.$$

Use the expressions

$$x = \frac{z + \overline{z}}{2} \qquad \text{and} \qquad y = \frac{z - \overline{z}}{2i}$$

to write $f(z)$ in terms of z, and simplify the result.

Solution. Using

$$x = \frac{z + \overline{z}}{2}, \qquad y = \frac{z - \overline{z}}{2i},$$

we obtain

$$x^2 - y^2 = \frac{z^2 + \overline{z}^2}{2}, \qquad -2y = i(z - \overline{z}),$$

and

$$i(2x - 2xy) = i(z + \overline{z}) - \frac{z^2 - \overline{z}^2}{2}.$$

Therefore

$$\begin{aligned} f(z) &= \frac{z^2 + \overline{z}^2}{2} + i(z - \overline{z}) + i(z + \overline{z}) - \frac{z^2 - \overline{z}^2}{2} \\ &= \overline{z}^2 + 2iz. \end{aligned}$$

Thus, for the function as printed,

$$\boxed{f(z) = \overline{z}^2 + 2iz}.$$

□

Remark. The printed answer in the textbook is $z^2 + 2iz$, which does not agree with the printed function. The direct algebra above gives $\overline{z}^2 + 2iz$.

Exercise 14.4. Write the function

$$f(z) = z + \frac{1}{z} \qquad (z \neq 0)$$

in the form $f(z) = u(r, \theta) + iv(r, \theta)$.

Solution. With $z = re^{i\theta}$,

$$\frac{1}{z} = \frac{1}{r} e^{-i\theta}.$$

Hence

$$\begin{aligned} f(z) &= r(\cos\theta + i\sin\theta) + \frac{1}{r}(\cos\theta - i\sin\theta) \\ &= \left(r + \frac{1}{r}\right)\cos\theta + i\left(r - \frac{1}{r}\right)\sin\theta. \end{aligned}$$

Therefore

$$\boxed{u(r, \theta) = \left(r + \frac{1}{r}\right)\cos\theta, \qquad v(r, \theta) = \left(r - \frac{1}{r}\right)\sin\theta}.$$

□